UNIVERSITY OF FRONTIER TECHNOLOGY, BANGLADESH

Department of Cyber Security Engineering

Project Proposal

Project Title : Phishsheild - Phishing Email Detection System
Course Name : Object Oriented Programming Language Sessional
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PROJECT PROPOSAL

1. Project Overview

Email is now a way for people and organizations to talk to each other in todays digital world.With this convenience comes risks, especially phishing attacks.These attacks use emails that try to look like they come from sources but their real aim is to trick users into giving away sensitive information such as passwords, bank details or personal data.

The PhishShield system helps detect phishing emails.It was developed using Java.The PhishShield system analyzes parts of an email like the sender, subject line, content and embedded links.Based on predefined rules it determines whether an email is safe or if it is suspicious.The goal of this project is to create an effective tool that helps users make better decisions while using email.

This tool aims to reduce the risk of phishing attacks and help users protect their information.The PhishShield system is designed to be a tool in the fight, against phishing attacks.

2. Problem Statement

Email systems are used by a lot of people.Many of them do not have good ways to stop phishing attacks.Most email services use filters and do not look closely at incoming messages. So phishing emails can easily get past these filters.Get into peoples inboxes.

People have a time figuring out if an email is real or fake.This is even harder when phishing emails look like they are from people we trust.A lot of people do not know what to look for in phishing emails so they are more likely to fall for them.

Some things like links,fake email addresses and strange content are not always checked properly by simple systems.This makes it more likely that people will share information without knowing it.

We need a system that can look at emails and find phishing attempts on its own.Email systems need this.Such a system would mean people do not have to decide on their own all the time and it would give us protection from cyber threats.This would help keep email systems safe. Email systems would be safer, with this kind of system.

3. Proposed Solution & Objectives

Proposed Solution

The PhishShield system is made to fix the problems with email systems that are already out there. This system is built using Java.It looks at the emails people get to see if someone is trying to trick them.It checks the parts of an email like who sent it what the email is about what the email says and the links inside the email.Then it decides if the email is okay or if it might be a problem.

If an email seems weird or might be bad the PhishShield system points it out.Tells the person who got the email.This helps people be careful when they get emails that might be suspicious.The PhishShield system is like a helper that watches emails to make sure they are safe.

Objectives

- To look at emails that come in and try to find phishing attempts
- To find out if the person who sent the email is who they say they are and if the email is trying to trick people
- To find links in emails that might be harmful or fake
- To warn people when they get an email that seems suspicious
- To help people know more, about phishing and make it less likely to happen to them

4. Target Audience

PhishShield is made for people who use email a lot and might get phishing emails.These emails can look real.It is easy to get tricked if you do not know much about technology.PhishShield helps find emails that might be fake and makes email safer.

Target Groups

General Email Users: These are people who use email every day to talk to friends and family. They need to be protected from emails that might hurt them.

Students: Students get a lot of emails, about school, scholarships and other things.Sometimes these emails are fake.Try to trick them.

Office Employees: People who work in offices get emails and need to be careful so they do not lose important information or get cheated.

Small Business Owners: These people use email to talk to customers and do business.They are often targeted by people who try to trick them.

Basic System Administrators: These are people who take care of email systems in companies.They need help to watch out for emails that might be suspicious.PhishShield is a tool for them to use to keep their email system safe.

5. Key Features & Project Scope

Key Features

1.Email analysis based on the senders details, subject line, email content and links it contains
2.Detection of fake email addresses
3.Identification of links that could be harmful or misleading
4.Basic scanning of email content to spot phishing indicators
5.A system that alerts users about suspicious emails
6.Classification of emails as safe or potentially harmful

Project Scope

This project aims to create a Java-based educational tool.It shows how email security can be improved with detection methods.The tool focuses on phishing detection.It does not offer enterprise-grade security.

The system serves as a prototype.It helps users understand how real-world email security tools work. The system can be expanded in the future.It can include features like machine learning or integration, with a database.

6. Functional & Non- Functional Requirements

Functional Requirements

- The system needs to let users put in or get email data so it can be looked at.
- The system has to take and look at important parts of emails like who sent it what it is about what it says and any links it has.
- It should find patterns that seem suspicious based on rules that we already have.
- The system needs to say if an email is safe or if it is phishing.
- When the system finds a phishing email it should send out a warning.
- The system has to keep the results of email analysis so we can look at them later.

Non Functional Requirements

- The system should be able to analyze emails without taking too long.
- It has to give us results every time.
- The system should be easy to use and not too complicated.
- The system needs to work every time we check emails.
- It should be easy to fix and make better in the future.
- The system has to handle data in a way that keeps it safe and organized like Email data, for the Email system.
- The Email system has to keep our Email data secure.

7. Technology Stack

Programming Language: C++

Data Handling & Structure

- Array and ArrayList for data handling.
- String Processing
- Simple file handling

Development Tools

- We will use IntelliJ IDEA or Eclipse as our Integrated Development Environment (IDE).
- JDK, which is Java Development Kit will be required.
- Notepad++ can be used for editing if needed.

Concept Used

- Object-Oriented Programming also known as OOP will be applied.
- Basic logic and validation rules will be used.
- We will also apply pattern checking techniques, for analyzing email.

8. System Architecture & UI Design

Input Module
This module gets email information like the senders address and subject.It also gets the message and any links that are in the email.The user can give this information.It can come from a list that is already stored.

Data Processing Module
This module looks at the email and takes out the important parts that need to be checked.

Detection Module
This is the part of the system.It checks emails to see if they are okay or not.It looks for things like sender addresses, weird words and bad links that might hurt the computer.

Classification Module
After the system checks the email this module decides if the email is Safe or if it is Phishing.

Alert Module
If the system finds something it sends a warning to the user right away.

System Flow

The system works like this:

Email goes in → Data Processing happens → Detection Module checks it → Classification decides what it is → Result is shown to the user

UI Design

The system has an interface that is easy to use.It is like a conversation where the user can type in the email information like who sent it what it is about what the message.Any links that are, in it. Then the system tells the user what it found out.If something bad is found the system warns the user.The system is made to be simple so anyone can use it.

9. Project Timeline & Milestones

Milestone	Description	Timeline
Milestone 1	Requirement analysis and project planning	Week 1
Milestone 2	System design and setting up the module structure	Week 2
Milestone 3	Develop the input and data processing modules	Week 3
Milestone 4	Implement the detection and classification logic	Week 4
Milestone 5	Testing,debugging and make improvements	Week 5
Milestone 6	Final documentation and preparing the project	Week 6

10. Conclusion

PhishShield is a system that helps find phishing emails.It looks at the sender, content and links, in an email to check if it is suspicious.This project makes users more aware of phishing attacks.It helps them not share information by mistake.PhishShield is a system.It shows how C++ programming can make email security better.In the future we can make PhishShield better.We can add machine learning and a database.This will make PhishShield more accurate and faster.